Edge Detection based Salient Region Detection for Accurate Image Forgery Detection

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***Abstract*— Image Forgery is a real time issue in this present era and that causes inimical effects in the society. This is more common due to the availability of numerous image modifications software. The ultimate aim of this project is to provide a best optimal solution to this existing problem. The forgery in images includes object removal, object addition, unusual color modifications. There are many existing techniques available to overcome this problem but they have some limitations such as lack of accuracy in saliency detection, larger hash values, deficient in detecting small area tampering. In this proposed method the image is preprocessed to fixed size so that generated hash value of fixed size. The resized image is converted into YCbCr image and is passed into Gaussian filter, blurred image is obtained. The saliency regions are accurately detected by using edge detection concepts, the global and local features are extracted from this detected region and the image is processed to remove the noise. The final image is obtained from which the object is extracted and edge pixels are detected and mapped to the original image, sensitive hash is constructed for those detected regions. This proposed method outperforms the existing system by accurately detecting saliency regions, reducing hash length and increases the sensitivity of the hash, so that even the small area tampering can be detected accurately.**

***Index Terms*—Edge detection, salient region detection, hash, tampering, reduced hash length, Image Authentication.**

I. Introduction

The image forgery becomes popular due to the increased image modification software. There are various Image manipulation software all available which may lead to tampering of images, preserving those image contents is very much essential. Image hash is the short sequence of message content. The hash must be very sensitive so that even the minute modification in the content will result in drastic difference in the value of hash. Eventually, the hash should be, reasonably short and so that it can reduce the complexity. The hash should provide Robustness to ordinary image processing (e.g., compression) and very sensitive to modification of contents. The key is generated and that should not be traced easily by the intruders. The security has to be ensured at this point. The accurate salient Regions are detected by means of edge detection from which global and local features are extracted (when we extract features by considering whole image it is global and when considering only part of the image it is local). The global features are basically short, but serious to changes where as local features are sensitive to regional changes, but produces longer hash. Here we combine both the local and global features. The main goal is to reduce the hash length, without affecting the performance of hash and to detect small area tampering. To achieve this, we concentrate on detecting accurate salient regions of an image. The local and global features are extracted and that can provide high sensitive hash code by using a technique called hash sensitivity growth method (HSGM), for image authentication. The image along with its encrypted hash is sent to the receiver and the receiver decomposes the hash and the distance is calculated between the Original image hash and the decrypted hash. If there is a change then it results in detecting and locating the forgery. The medical image, Military image, Quality control and the evidence in the courts are some of the documents that describe the importance of detecting the forgery. The tamper in these areas are not tolerated. An example of an original image and modified Image is given in the Fig 1 and 2.

Figure 1) Original Image

Figure 2) Tampered Image

The Figure 1 shows the original image and the image is modified as shown in Figure 2. The Original image is modified (2 and 3 are modified as 6 and 7). The local texture features and the Characteristics are extracted from salient regions in the image to represent contents in the corresponding area. The degree of similarity between two generated hashes is defined to measure the hash performance by the distance metrics. Whether a given image is an original image or maliciously modified or totally a different image is calculated using two thresholds. The proposed method is capable of locating the area of tampering and classifying the nature of tampering, e.g., replacement of objects or abnormal

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modification of colors. The remaining sections of paper are organized as follows. In Section II Related Works are described. In section III salient region detection, feature extraction and edge detection are described. Section IV the proposed image hashing scheme and the procedure of image authentication are discussed. Section V concludes the paper. Section VI describes the future enhancements.

II. RELATED WORKS

In most of the existing methods as in [8] they use the watermarking based approaches where the information is attached in the LSB bits of the image. There are two types of watermark, visible watermarking and invisible watermarking. One of the major disadvantages of watermarking is that it will deform the content. In order to overcome this, hashing was introduced. The hashing technique becomes more sensitive even to normal image processing and hence Robust hashing becomes more popular and it is used in many existing methods as Robust hashing for similarity search [2], A clustering based approach to perceptual image hashing [3], Robust image hashing for tamper detection using non-negative matrix factorization [4], Robust and secure image hashing [5], and all these hash are signature based methods. In these methods, a hash code is entrenched with the image before transmission. At the destination the hash code is decomposed and used for verification. In [1] the saliency regions are detected by means of rectangle shaped structure. The image hash is the signature which represents the visual content. The hash of the image should be robust to normal image processing and sensitive to tampering. The other latest researches on this accurate saliency detection are [8] and [9], which are exploiting the benefits from both the spatial domain and the spectral domain for saliency. The proposed method provides systematic evidence against the malicious manipulations. The main part of this paper is to accurately detect the saliency regions of an image and to extract the features, both local and global, by using the edge detection concept the edge pixels of the features are mapped to original image and thus increases the sensitivity of the hash by which it could detect small area tampering.

III. PROPOSED STRATAGEM

The stratagem for the proposed hash scheme and process involved in the image authentication are as follows

A. Coining the Hash

Hash is the short sequence that represents the character of the image and it is used for image authentication. Generating the image hash involves various procedures, they are

- Image Pre-processing
- Saliency Detection
- Saliency Detection Algorithm
- Global features extraction
- Local features extraction
- Edge Detection for Hash Construction

Figure 3 shows the proposed stratagem block diagram.

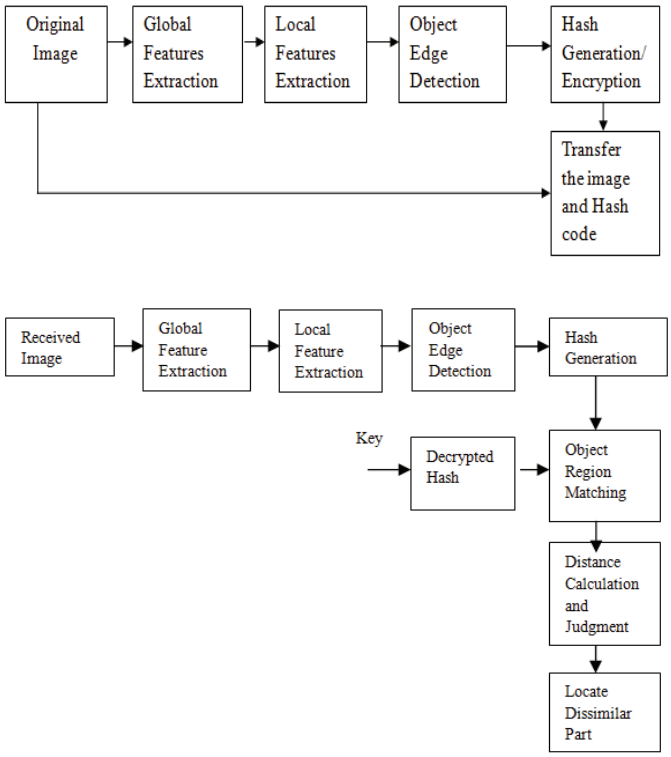

Figure 3) Block diagram of proposed stratagem

a) Pre-processing the image: The input, original image is re-sized into fixed size image, and further image RGB is converted to the YCbCr representation as in figure 4 and 5.

Figure 4) RGB

Figure 5) YCbCr

b) Salient Region Detection: Saliency is strong-minded as the local contrast of an image region with respect to its neighborhood at various scales. This is evaluated as the distance between the average feature vectors of the pixels of an image sub-region with the average feature vector of the pixels of its neighborhood as in [7] is shown in figure 6.

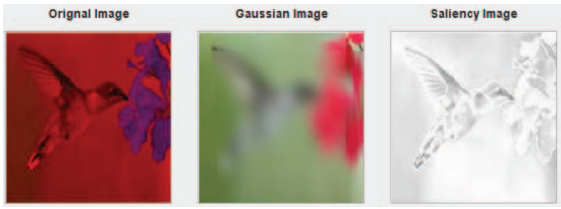

Figure 6) Saliency Region Detection

c) Saliency Detection Algorithm: This algorithm is used to detect saliency regions by finding the Euclidean distance

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between the *Lab* pixel vector in a Gaussian filtered image with the average *Lab* vector for the input image [7]. The Figure 7 describes the process involved in obtaining the saliency image.

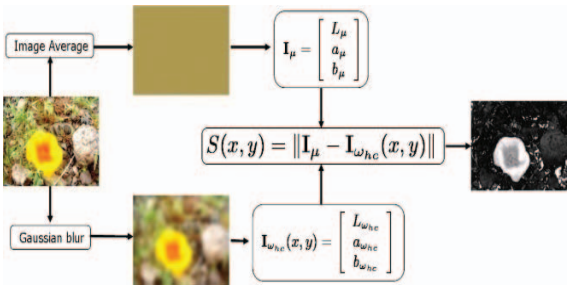

Figure 7) Process of Saliency Detection

d) Global Feature Extraction: When considering the whole image it is global and for the extraction calculates the Y Cb and Cr using the Zernike moments to form global feature.

e) Local Feature Extraction: When considering only the part of the image (Position and texture of salient regions) it is local feature. The Salient regions include the coarseness and contrast of the image. The feature extraction is showed in figure 8.

Figure 8) Feature Extraction

f) Edge Detection for Hash Construction: The features are extracted from the saliency regions and edge detection concept is applied here, by which the edge pixels of the features are detected and mapped to the original image as in figure 10. The global and salient local feature vectors are concatenated to obtain an intermediate hash and that is then pseudo-randomly jumbled based on a secret key to produce the final hash sequence of fixed length and also with high sensitivity growth.

B. Proposed Framework for Hash Construction

In [2] the author proposed robust hashing with local models for similarity search and proved it to be efficient. Our idea is to use this concept for Image Authentication for generating sensitive hash values. Considering local features alone is enough for similarity search but for authentication we consider both local and global features and proposed a technique called Hash Sensitivity Growth Model (HSGM). The proposed framework, used for generating the Hash values has two phases.

- Training Phase
- Testing Phase

a) Training Phase: This phase is offline, it is used to learn hash functions effectively, which is used to map data points located in the testing database into binary hash codes. It has two main components,

- learning of hash codes
- learning of hash functions

If a training dataset is given, we first learn the hash codes of the training dataset using our proposed model. Then, in the learning of the hash function component, by considering both the training data and their learned hash codes, a series of hash functions are learned. Each hash function can be used to generate one bit of the hash code for a data point. By specifying the length of the hash code *l*, different sized binary hash codes can be generated.

b) Testing Phase: At a query point, performing a similarity search is quite simple. First, the given query is mapped into its binary hash code as explained above and, followed by searching the hash table.

C. Comparison between rectangular saliency and edge detection based saliency detection

When the saliency is detected by the existing method then it detects the saliency by means of rectangular shape and hence the hash value will be large. In the proposed concept they use edge detection based technique so it detects the edges of the extracted feature and the saliency is accurate hence only the edges of the image features detected are extracted and mapped with the original image, hence sensitivity of the hash is increased and also the noises are removed. This edge detection technique accurately detects saliency and sensitive to small area tampering as the hash sensitivity is increased.

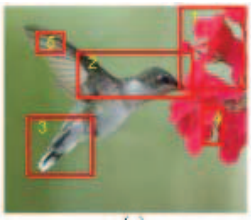

Figure 9) Existing Method (Rectangular Detection)

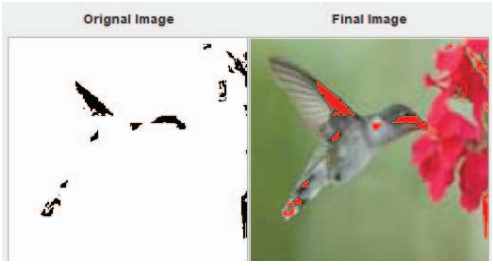

Figure 10) Proposed Method (Edge Detection)

From the result in figure 9 the saliency is detected with four rectangles and this decreases hash sensitivity and also produces longer hash but by proposed method as in figure 10 the saliency detected accurately and edge pixels are detected by using the edge detection and mapped to original image, hence red color indicates features and, for this alone hash is

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constructed so the value of hash will be small. Hence the proposed method outperforms the contemporary methods available, comparatively and reduces complexity.

D. Image Authentication

In this image authentication, the receiver decomposes the hash value and compares the hash value with that of the original image and if they both match then it determines the test image is similar to the reference image. The image authentication undergoes the following steps

- Extraction of Features
- Decomposing the Hash Value
- Matching the Saliency Regions

a) Extraction of Features: The received image is verified by the same procedures as above, except the encryption. The features are extracted from the image

b) Decomposing the Hash Value: The intermediate hash code is restored using the secret key from the reference hash, which is a concatenated feature sequence of the original image. It is decomposed into global and local features.

c) Matching Saliency Region: Whether, the salient regions detected in the received image are similar to that of the original image is checked.

This process is the reverse process of coining the hash. The hash value is decomposed and the features are extracted from which salient regions are detected and matched.

E. Hamming distance computation

The Hamming distance between two binary hash codes are computed by the very much proficient XOR operation. The distance between the received and original image is calculated so as to identify and locate the forgery in an image.

F. Forgery Classification and Localization

The Hamming distance algorithm is used to find the distance between the original and a test image. Based on the difference, image forgery is identified in four ways as follows,

- Object added
- Object removed
- Unusual Modification of Color
- Object Misplaced

IV. CONCLUSION

In this paper, the robust hashing with local model along with global features combined and a new hashing stratagem is proposed and the new method Hash Sensitivity Growth Model (HSGM) overcomes the drawbacks of the existing image authentications problems. The proposed method preserves the local structural information by a novel local hashing models constructed for each individual data point in the training data set. This produces effective optimized image forgery detection by increasing the sensitivity of the hash by accurately detecting the saliency regions. The accurate saliency region detection is obtained by edge detection and extraction of both local and global features produces highly sensitive hash which can detect even small area tampering. It can proficiently detect forgery and robust to normal image processing. It preserves the contents and local structures of an image and overcomes the drawback of locality sensitive hashing.

V. FUTURE ENHANCEMENT

With this proposed hashing scheme, accurate salient regions are detected by means of edge detection technique, from that both local and global features are extracted to construct the hash. This preserves the local structure content and hence very sensitive to small area tampering due to increased hash sensitivity. The future enhancement of this will focus on IP tracer routing.

REFERENCES

[1] Yan Zhao, Shuozhong Wang, Xinpeng Zhang, and Heng Yao, Member, IEEE, ”Robust Hashing for Image Authentication Using Zernike Moments and Local Features”, IEEE Transactions on Information Forensics and Security, vol. 8, no. 1, January 2013.

[2] Jingkuan Song, Yi Yang, Xuelong Li, Fellow, IEEE, Zi Huang, and Yang Yang, ”Robust Hashing With Local Models for Approximate Similarity Search”, IEEE Transactions on Cybernetics, vol. 44, no. 7, July 2014 1225

[3] V. Monga, A. Banerjee, and B. L. Evans, “A clustering based approach to perceptual image hashing,” IEEE Trans. Inf. Forensics Security, vol. 1, no. 1, pp. 68–79, Mar. 2006.

[4] Z. Tang, S.Wang,X. Zhang, W.Wei, and S. Su, “Robust image hashing for tamper detection using non-negative matrix factorization,” J. Ubiquitous Convergence Technol., vol. 2, no. 1, pp. 18–26, May 2008.

[5] Swaminathan, Y. Mao, and M. Wu, “Robust and secure image hashing,” IEEE Trans. Inf. Forensics Security, vol. 1, no. 2, pp. 215–230, Jun. 2006.

[6] F. Khelifi and J. Jiang, “Perceptual image hashing based on virtual watermark detection”, IEEE Trans. Image Process., vol.19, no.4, pp.981–994, Apr.2010.

[7] R. Achanta, S. Hemami, F. Estrada and S.Süsstrunk, “Frequency-tuned Salient Region Detection”, IEEE International Conference on Computer Vision and Pattern Recognition (CVPR 2009), pp. 1597 - 1604, 2009.

[8] Meenakshi Sundaram A. and C. Nandini, “Feature based Image Authentication using Symmetric Surround Saliency Mapping in Image Forensics” International Journal of Computer Applications (0975 – 8887) Volume 104 – No.13, October 2014.

[9] Hong-Yun Gao and Kin-Man Lam, “Saliency detection based on adaptive DOG and distance transform”, IEEE International Symposium Circuits and System (ISCAS), June 2014.

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